The Pitch Count Battle

Do teams that throw fewer pitches win more, and why?

Trent Reimers

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The Question

Throughout the 2026 UCLA season, I often joked around in the manager's office about winning the "pitch-count battle" early in games when it was 0-0 or even down. I firmly believed in this idea and felt it worked more often than not for the Bruins. Now that I am out of the office, I wanted to know how much of that is true: **do teams that throw fewer pitches actually win more, and if so, why?**

I looked at 14,505 MLB regular-season games from 2021 to 2026 (Statcast) and 6,471 NCAA Division I games from 2026 (TrackMan). I looked at the full game, and at the pitch battle through 3 and through 5 innings.

Method

Normalize per 27 outs, not per game. When the home team wins, the road team rarely pitches the bottom of the final inning. A raw pitch count gives the losing road team credit for an inning it never had to pitch or complete. Every team-game is scaled to pitches per 27 outs recorded. That also handles walk-offs, extra innings, and 7-inning college games.

Split the pitch count into its parts. Pitches per 27 outs breaks down exactly:

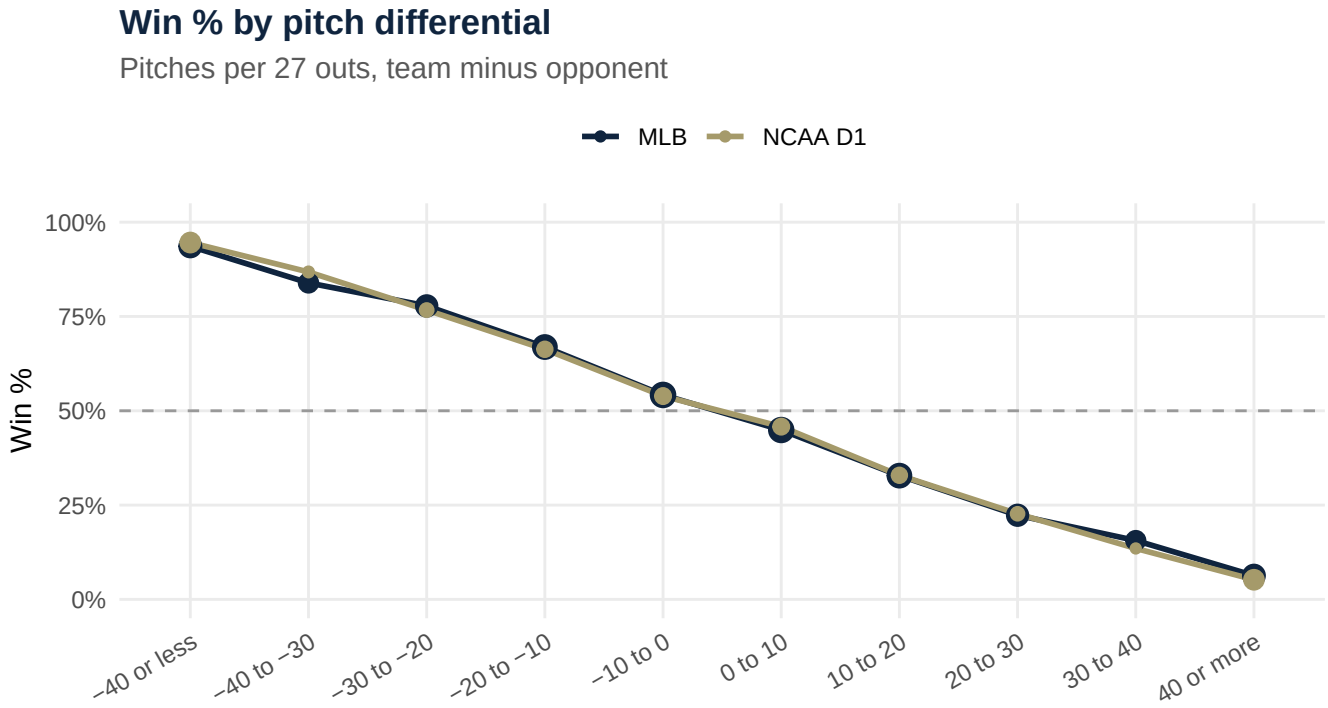
$$\frac{\text{Pitches}}{27\text{ outs}} = 27 \times \underbrace{\frac{\text{PA}}{\text{out}}}_{\text{baserunners}} \times \underbrace{\frac{\text{Pitches}}{\text{PA}}}_{\text{count battle}}$$

The first part is how many hitters you let reach. The second is the "count battle" in its pure form: how long each plate appearance lasts.

Control for strikeouts and damage. Strikeouts take more pitches by definition, and long PAs could simply mean less hard contact. So I added strikeout rate and contact damage (extra bases per ball in play, so it doesn't double-count hits already in the baserunner term). I also measured the count battle using only PAs that *didn't* end in a strikeout. Every variable is the team's value minus the opponent's, and each game counts once. Effects are reported in

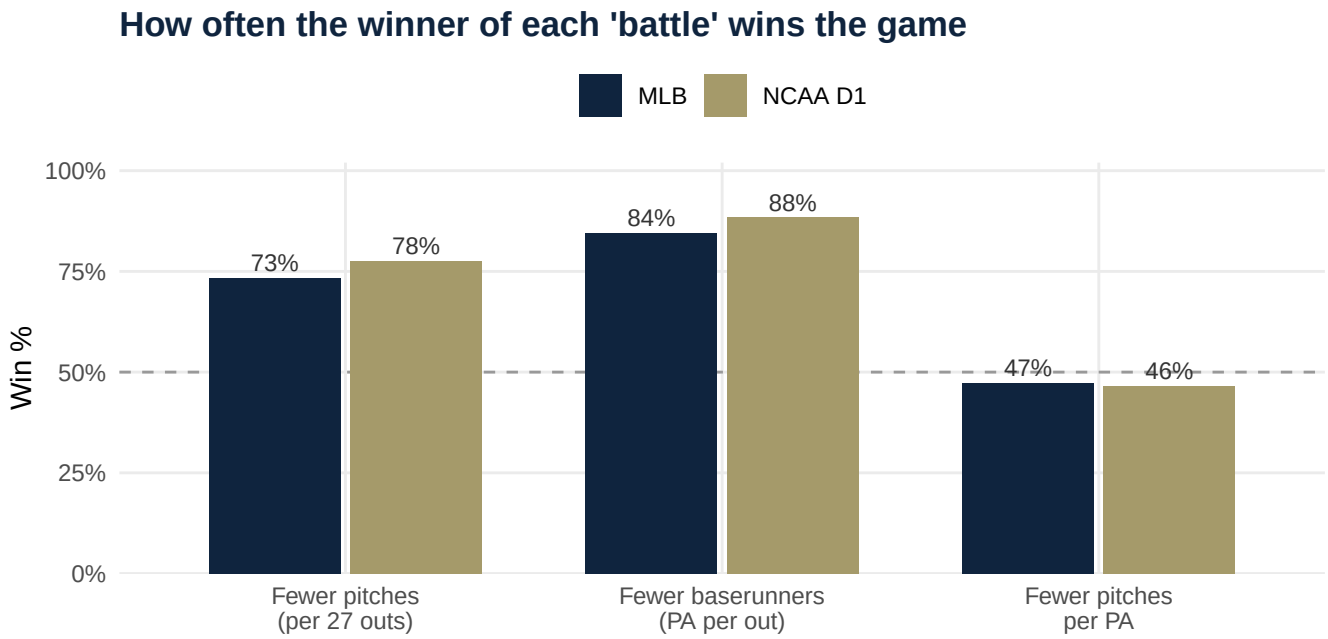
win percentage per standard deviation, from a linear probability model with robust errors and checked against logistic average marginal effects.

Result 1: Fewer Pitches Wins



The curve is close to a straight line, and it's nearly identical in MLB and college. The team that threw fewer pitches per 27 outs won **73.4%** of MLB games and **77.6%** of D1 games. The raw per-game count understates this (61.2% and 66.6%) because of the road-loser bias.

Result 2: It's about Baserunners, not PA Efficiency



Split the pitch count into its two parts and the story changes. Allowing fewer baserunners wins **84.5%** (MLB) and **88.4%** (D1) of the time. Throwing fewer pitches *per plate appearance* wins less than half the time (47.4% and 46.5%). Quick, efficient PAs are not what separates winners. Low pitch counts are mostly a side effect of keeping runners off base.

Result 3: What's Left of the Count Battle?

Holding baserunners equal, the team with *longer* PAs actually wins more. Some of that is strikeouts. The table shows how much of the original count-battle effect survives each control.

Table 1: Count-battle effect as a share of the baseline (logistic models)

Model	MLB	NCAA D1
Baserunners + count battle (baseline)	100%	100%
+ strikeout rate	66%	75%
+ damage	127%	122%
+ strikeout rate + damage	58%	70%
Non-K count battle + K rate + damage	66%	74%

Strikeouts absorb 34% (MLB) and 25% (D1) of it. Damage absorbs none, because longer PAs aren't linked to softer contact. In log-odds, about 66% (MLB) and 74% (D1) survives. Log-odds can overstate how much that matters, though, so the real test is win percentage.

Result 4: In Win Percentage, It's Small

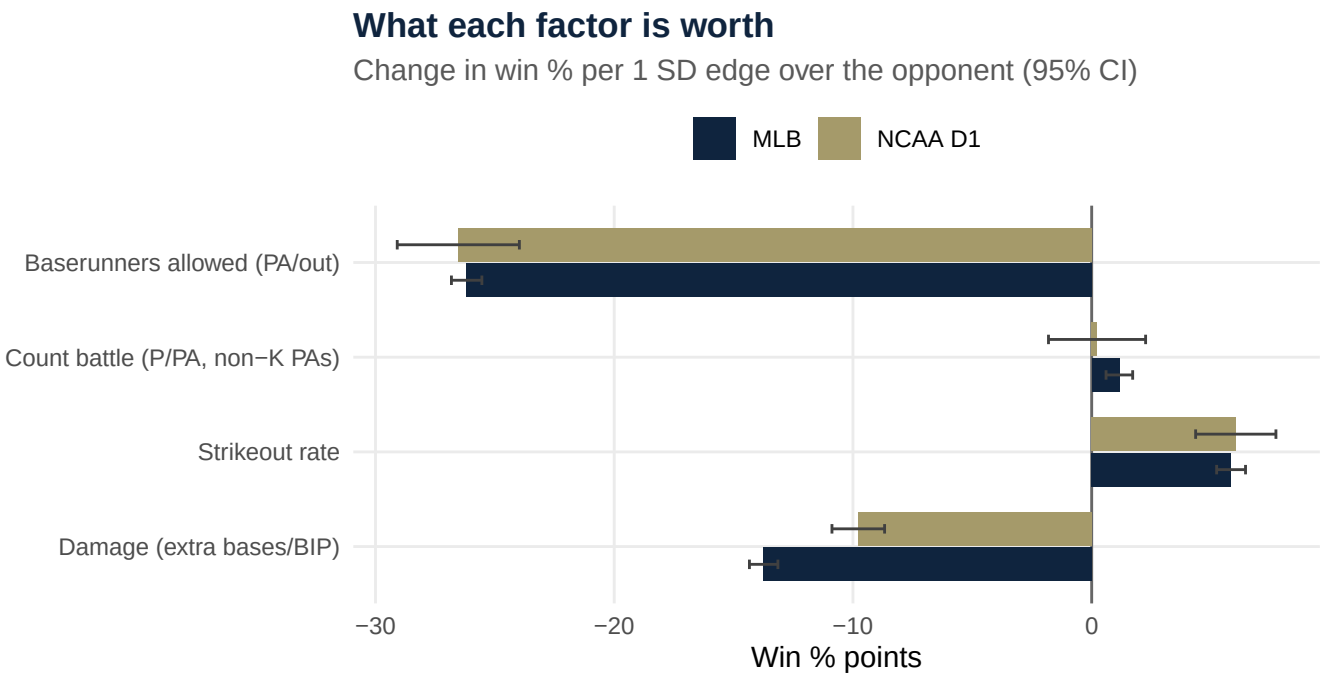


Table 2: Win % points per SD: linear probability model and logistic average marginal effect

Factor	MLB	MLB (logit AME)	D1	D1 (logit AME)
Baserunners allowed (PA/out)	-26.2	-25.4	-26.5	-29.4
Count battle (P/PA, non-K PAs)	+1.2	+1.2	+0.2	+1.6
Strikeout rate	+5.8	+5.0	+6.0	+4.4
Damage (extra bases/BIP)	-13.7	-13.4	-9.8	-9.9

Allowing fewer baserunners is worth about **26 points** of win % per standard deviation at both levels. Damage is next (-13.7 MLB, -9.8 D1), then strikeouts (+5.8 and +6.0). The pure count battle is worth **+1.16 points in MLB** ($p = 4.7 \times 10^{-5}$) and **+0.22 in D1** ($p = 0.83$).

Result 5: The Office Theory

This is the version of the question I actually argued about in the office: if you're tied or down early, does being ahead in the pitch count still mean something? For each game, I froze the score and pitch counts after 3 and after 5 complete innings, then asked how often the team that had thrown fewer pitches went on to win, at each score.

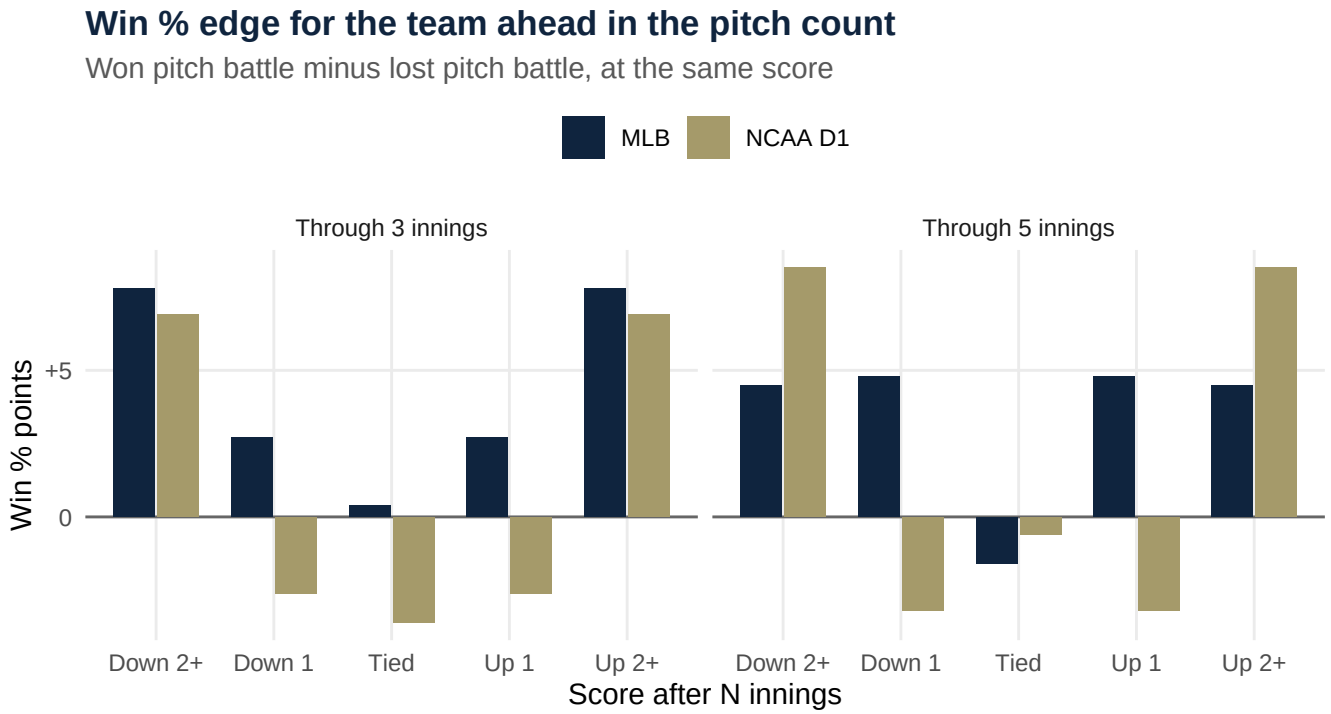


Table 3: Win % for the team ahead in the pitch count after N innings

Level	Through	Games	All games	Tied games	Per 10 fewer*
MLB	3 inn.	14,505	62.9%	50.2%	+0.8
MLB	5 inn.	14,505	66.7%	49.2%	+0.5
NCAA D1	3 inn.	6,219	65.1%	48.2%	+0.4
NCAA D1	5 inn.	6,114	70.8%	49.7%	+0.4

*Win % points per 10 fewer pitches than the opponent, score held fixed.

Being ahead in the pitch count after 3 innings goes with winning 62.9% of MLB games and 65.1% of D1 games. After 5, it's 66.7% and 70.8%. Most of that is just the scoreboard, though. Teams with fewer pitches early are usually the ones ahead.

The fairer test is the one from the manager's office: same score, different pitch counts. **When it's tied after 3, the team ahead in the pitch count wins 50.2% (MLB) and 48.2% (D1). That's a coin flip.** Down a run after 3, winning the pitch battle is worth +2.7 points in MLB and -2.6 in D1, so it doesn't even point the same way at both levels. Holding the score fixed, every 10 fewer pitches through 5 innings is worth +0.5 points of win % in MLB and +0.4

in D1.

The only bars that clear 5 points are in games already two or more runs apart. At the scores the office theory was actually about, tied or down one, the early pitch battle tells you almost nothing about who wins.

Takeaways

1. **Fewer pitches really does go with winning.** The edge is steep and almost the same in MLB and college.
2. **But the pitch count is a symptom.** Nearly all of it comes from allowing fewer baserunners and less damage. "Pitch to contact and save pitches" is the wrong takeaway from this. The real lesson is run prevention: keep hitters off base and don't give up extra-base hits.
3. **The count battle itself is a small MLB edge and close to nothing in college.** With the same baserunners, strikeouts, and power allowed, longer non-strikeout PAs are worth about 1.2 points of win % per standard deviation in MLB. A plausible explanation is starter fatigue and earlier bullpen exposure, which matter more in a league built around five-man rotations and deep bullpens.
4. **The office theory doesn't hold up.** Tied after 3, the team ahead in the pitch count battle wins 50.2% in MLB and 48.2% in D1. An early pitch-count lead mostly tells you who is already winning on the scoreboard. With the score even, it doesn't predict the rest of the game.

Data notes. MLB: 2021–2026 regular seasons, Baseball Savant via `baseballr`. NCAA: 2026 Division I TrackMan files, de-duplicated by pitch ID. Games with tied TrackMan scores or fewer than 18 outs or 60 pitches for either side were removed. Outs come from TrackMan out and strikeout fields, and from Statcast half-innings, with walk-off final halves counted partially. Damage = (2B + 2×3B + 3×HR) / balls in play. All differentials are team minus opponent, z-scored within level. Through-N splits use only games where both teams completed N full innings, with the score held fixed using run-differential indicators capped at ±4.